Chemistry - Undergraduate Level Assessment 23 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which of the following metal ions cannot be used as a paramagnetic quencher?
   (a) Ti_3+
   (b) Cr_3+
   (c) Fe_3+
   (d) Zn_2+

2. What is the orbital angular momentum quantum number, l, of the electron that is most easily removed when ground-state aluminum is ionized?
   (a) 3
   (b) 2
   (c) 1
   (d) 0

3. Of the following ions, which has the smallest radius?
   (a) K+
   (b) Ca_2+
   (c) Sc_3+
   (d) Rb+

4. Which of the following types of spectroscopy is a light-scattering technique?
   (a) Nuclear magnetic resonance
   (b) Infrared
   (c) Raman
   (d) Ultraviolet-visible

5. Infrared (IR) spectroscopy is useful for determining certain aspects of the structure of organic molecules because
   (a) all molecular bonds absorb IR radiation
   (b) IR peak intensities are related to molecular mass
   (c) most organic functional groups absorb in a characteristic region of the IR spectrum
   (d) each element absorbs at a characteristic wavelength

True / False (10 marks)

Answer True or False

6. True or False: The atomic radii of the lanthanide elements decrease across the period from La to Lu due to increasing nuclear charge.

7. True or False: In the lowest energy state of a one-dimensional box, momentum can be known exactly while position remains uncertain.

8. True or False: The magnetic field responsible for the polarization of 2.5 x $10^{-6}for13Cat298Kis2.9T$.

8. True or False: In a three-component mixture, a maximum of five phases can coexist at equilibrium with each other.

9. True or False: NaCl has a greater lattice enthalpy than MgO because sodium and chloride ions are larger than magnesium and oxide ions.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the significance of the R-branch in the vibrational-rotational spectrum of a diatomic molecule, focusing on the specific transitions involved and their implications for molecular behavior.

12. Calculate the NMR frequency of 31 P in a 20.0 T magnetic field and discuss the factors influencing this frequency in nuclear magnetic resonance spectroscopy.

End of Paper